LMSR as Entropy-Smoothed Batch Auction Clearing

Proof Sketch

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Summary. We prove that Hanson's Logarithmic Market Scoring Rule (LMSR) is the entropy-smoothed version of a welfare-maximizing batch auction with minting. The connection is exact: the LP's minting cost function $V(D) = \max_k D_k$ and the LMSR cost function $C_b(D) = b \cdot \ln \sum_k \exp(D_k/b)$ are related by $C_b \to V$ as $b \to 0$. Their Fenchel conjugates exhibit the same convergence: the negative entropy $-b \cdot H(p)$ converges to the simplex indicator $\delta_\Delta(p)$. This establishes that LP batch clearing and LMSR pricing are endpoints of a single parametric family, and that the LP inherits all structural properties of LMSR (price normalization, arbitrage-freeness) as limiting cases.

1. The Minting Cost Function

1.1. Setup

Consider a prediction market with K mutually exclusive outcomes. In a batch auction, N orders arrive. After matching buyers against sellers, the residual imbalance is:

$$D_k = \sum_{i \in \text{ buy}(k)} q_i - \sum_{j \in \text{ sell}(k)} q_j \quad \text{for each outcome } k \in \{1, ..., K\}$$

where q_i is the fill quantity of order i. The vector $\boldsymbol{D} = (D_1, ..., D_K)$ is the *net demand* — the excess of buy volume over sell volume for each outcome.

To clear the market, the exchange must provide D_k additional shares of outcome k. Since outcomes are mutually exclusive, one *group mint* at cost \$1 creates one share of every outcome (exactly one resolves to \$1 at settlement, so this is fairly priced).

1.2. LP Minting Cost

The minimum minting cost to satisfy demand $\boldsymbol{D}$ is:

Definition 1 (Minting Cost Function).

$$V(\boldsymbol{D}) = \max_k D_k$$

Justification. Each group mint creates one share of every outcome. To supply D_k shares of outcome k for all k, we need at least $\max_k D_k$ mints (each mint covers one unit of every outcome). The cost is \$1 per mint, so the total cost is $\max_k D_k$. If $\max_k D_k < 0$ (all outcomes have excess supply), we *burn* pairs and recover revenue. ◇

The function $V = \max_k D_k$ is convex, piecewise linear, and non-smooth (it has kinks where $D_j = D_k$ for distinct j, k).

1.3. The Full Batch Auction as an Optimization

The welfare-maximizing batch clearing solves:

$$P : \max_{\boldsymbol{q} \in [0, \overline{\boldsymbol{Q}}]} \sum_i w_i q_i - V(\boldsymbol{D}(\boldsymbol{q}))$$

where w_i is the welfare coefficient of order i ($+L_i$ for buyers, $-L_i$ for sellers), and $\boldsymbol{D}(\boldsymbol{q})$ is the net demand vector (linear in $\boldsymbol{q}$). Since V is convex and $\boldsymbol{D}$ is linear in $\boldsymbol{q}$, the objective is concave — this is a convex optimization problem.

Introducing an explicit minting variable $\mu \geq \max_k D_k$, this is equivalent to the Linear Program:

$$\max_{\boldsymbol{q},\mu} \quad \sum_i w_i q_i - \mu \quad \text{s.t.} \quad D_k(\boldsymbol{q}) \leq \mu \quad \forall k, \quad \boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]$$

2. Fenchel Duality: Prices from Conjugates

The structural properties of the LP clearing prices follow from the Fenchel conjugate of V.

2.1. The Conjugate of V

Theorem 1 (Minting–Simplex Duality). *The Fenchel conjugate of $V(\boldsymbol{D}) = \max_k D_k$ is the indicator function of the probability simplex:*

$$V^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \left\{ \langle \boldsymbol{p}, \boldsymbol{D} \rangle - \max_k D_k \right\} = \delta_\Delta(\boldsymbol{p})$$

where $\Delta = \left\{ \boldsymbol{p} \in \mathbb{R}^K : p_k \geq 0, \sum_k p_k = 1 \right\}$ *and* $\delta_\Delta(\boldsymbol{p}) = 0$ *if* $\boldsymbol{p} \in \Delta$, $+\infty$ *otherwise.*

Proof. We compute $V^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \left\{ \sum_k p_k D_k - \max_k D_k \right\}$ in three cases.

Case 1: $\boldsymbol{p} \in \Delta$. For any $\boldsymbol{D}$, we have $\sum_k p_k D_k \leq \max_k D_k$ (a convex combination of the D_k cannot exceed the maximum). So the supremum is ≤ 0. Setting $\boldsymbol{D} = \boldsymbol{0}$ gives value 0. Therefore $V^*(\boldsymbol{p}) = 0$.

Case 2: $\sum_k p_k > 1$. Take $\boldsymbol{D} = (t, t, ..., t)$ for $t \to +\infty$. Then $\sum p_k t - t = t(\sum p_k - 1) \to +\infty$.

Case 3: Some $p_j < 0$. Take $D_j \to -\infty$ with all other $D_k = 0$. Then $p_j D_j \to +\infty$ while $\max_k D_k = 0$, so the supremum diverges.

In Cases 2 and 3, $V^*(\boldsymbol{p}) = +\infty$. □

Interpretation. The batch auction's minting mechanism *is* the probability simplex. The constraint that prices must form a valid probability distribution is not imposed externally — it is the Fenchel conjugate of the minting cost. Minting at \$1 per complete set *encodes* the axiom $\sum p_k = 1$.

3. Log-Sum-Exp Smoothing

3.1. The LMSR Cost Function

Definition 2 (Smoothed Minting Cost). *For temperature parameter $b > 0$:*

$$C_b(\boldsymbol{D}) = b \cdot \ln \sum_{k=1}^{K} \exp\left(\frac{D_k}{b}\right)$$

This is the *log-sum-exp* (LSE) function, scaled by b. It is convex, smooth (C^∞), and approximates V:

Proposition 1 (LSE–Max Sandwich). *For all $\boldsymbol{D} \in \mathbb{R}^K$:*

$$\max_k D_k \leq C_b(\boldsymbol{D}) \leq \max_k D_k + b \ln K$$

In particular, $C_b(\boldsymbol{D}) \to V(\boldsymbol{D})$ uniformly on compact sets as $b \to 0^+$.

Proof. Lower bound: $\sum \exp(D_k/b) \geq \exp(\max_k D_k/b)$, so $C_b \geq \max_k D_k$. Upper bound: $\sum \exp(D_k/b) \leq K \exp(\max_k D_k/b)$, so $C_b \leq \max_k D_k + b \ln K$. □

The gap $b \ln K$ is precisely the maximum loss of the LMSR market maker — a well-known result that here falls out as an approximation bound.

3.2. The Conjugate of C_b

Theorem 2 (LMSR–Entropy Duality). *The Fenchel conjugate of C_b is the negative Shannon entropy, restricted to the simplex:*

$$C_b^*(\boldsymbol{p}) = \begin{cases} -b \cdot H(\boldsymbol{p}) = b \sum_k p_k \ln p_k & \text{if } \boldsymbol{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

where $H(\boldsymbol{p}) = -\sum_k p_k \ln p_k$ is the Shannon entropy.

Proof. We compute $C_b^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \{\langle \boldsymbol{p}, \boldsymbol{D} \rangle - b \ln \sum \exp(D_k/b)\}$.

Setting the gradient to zero:

$$\frac{\partial}{\partial D_k} \left[p_k D_k - b \ln \sum_j \exp\bigl(D_j/b\bigr) \right] = p_k - \frac{\exp(D_k/b)}{\sum_j \exp\bigl(D_j/b\bigr)} = 0$$

This gives $p_k = \exp(D_k/b) / \sum \exp\bigl(D_j/b\bigr)$ — the *softmax*, which is exactly the **LMSR marginal price**. Inverting: $D_k = b \ln p_k + b \ln Z$ where $Z = \sum \exp\bigl(D_j/b\bigr)$.

Substituting back:

$$\begin{aligned} \langle \boldsymbol{p}, \boldsymbol{D} \rangle &= \sum_k p_k (b \ln p_k + b \ln Z) = b \sum_k p_k \ln p_k + b \ln Z \\ C_b(\boldsymbol{D}) &= b \ln Z \end{aligned}$$

Therefore $C_b^*(\boldsymbol{p}) = b \sum_k p_k \ln p_k + b \ln Z - b \ln Z = b \sum_k p_k \ln p_k$.

This is finite only when $\boldsymbol{p} \in \Delta$ (the softmax always yields a probability vector). □

3.3. The Duality Diagram

The two pairs (V, V^*) and (C_b, C_b^*) form a commutative diagram under Fenchel conjugation and the $b \to 0$ limit:

$$\begin{array}{ccc} V(\boldsymbol{D}) = \max_k D_k & \xleftrightarrow{\text{Fenchel}} & V^*(\boldsymbol{p}) = \delta_\Delta(\boldsymbol{p}) \\ \uparrow\ b \to 0 & & \uparrow\ b \to 0 \\ C_b(\boldsymbol{D}) = b \ln \sum \exp(D_k/b) & \xleftrightarrow{\text{Fenchel}} & C_b^*(\boldsymbol{p}) = b \sum p_k \ln p_k \end{array}$$

Left column: **primal** (cost of minting shares). Right column: **dual** (constraint on prices).
Top row: **LP batch auction** (sharp). Bottom row: **LMSR** (smooth, temperature b).

As $b \to 0$: the smooth LMSR cost C_b sharpens to the LP minting cost V, and the soft entropy penalty $b \sum p_k \ln p_k$ hardens to the simplex indicator δ_Δ (a rigid constraint that prices must be probabilities).

4. LMSR Pricing as Smoothed Batch Clearing

4.1. The Smoothed Batch Auction

Replace the minting cost V with C_b in the batch clearing problem:

$$P_b : \max_{\boldsymbol{q} \in [0, \overline{\boldsymbol{Q}}]} \sum_i w_i q_i - C_b(\boldsymbol{D}(\boldsymbol{q}))$$

Since C_b is convex and smooth, and $\boldsymbol{D}(\boldsymbol{q})$ is linear, P_b is a smooth concave maximization. Its first-order conditions are necessary and sufficient.

Theorem 3 (Main Result). *At the optimum of P_b, the marginal cost of shares — i.e., the clearing prices — are:*

$$p_k^* = \frac{\partial C_b}{\partial D_k} = \frac{\exp(D_k^*/b)}{\sum_j \exp\left(D_j^*/b\right)}$$

This is the LMSR marginal price function.

Furthermore:
1. $\sum_k p_k^* = 1$ *(prices form a probability distribution).*
2. *As $b \to 0^+$, the solution of P_b converges to the solution of P (the LP batch auction).*
3. *As $b \to \infty$, prices converge to the uniform distribution $p_k = 1/K$.*

Proof of (1). The softmax always sums to 1 by construction.

Proof of (2). By Proposition 1, the objectives of P_b and P satisfy $|\mathrm{obj}(P_b) - \mathrm{obj}(P)| \leq b \ln K \to 0$. The feasible sets are identical ($\boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]$). By standard results in epi-convergence of convex functions, the optimizers converge.

Proof of (3). As $b \to \infty$, $\exp(D_k/b) \to 1$ for all k, so $p_k \to 1/K$. □

4.2. Interpretation

The parameter b interpolates between two extremes:

- **$b = 0$ (LP batch auction):** Prices are sharp. The clearing price is set by the marginal order. Supply from minting is a step function: unlimited at \$1 per complete set, zero otherwise. Prices satisfy $\sum p_k = 1$ as a *hard* constraint (from $V^* = \delta_\Delta$).
- **$b > 0$ (LMSR):** Prices are smooth. The market maker provides a continuous supply curve with depth proportional to b. Prices satisfy $\sum p_k = 1$ by *softmax structure* (from $C_b^* = -bH$). The maximum loss from smoothing is $b \ln K$ (the entropy gap from Proposition 1).

The batch auction is the *zero-temperature* limit. Raising b adds liquidity (smoothness) at the cost of a bounded subsidy.

5. The KKT Conditions: Where the Exponentials Come From

To see exactly how LMSR pricing emerges from the optimality conditions, we write out the KKT system for P_b.

5.1. First-Order Conditions

For a buy order i on outcome k with welfare coefficient $w_i = L_i$:

$$\frac{\partial}{\partial q_i}[w_i q_i - C_b(\boldsymbol{D}(\boldsymbol{q}))] = L_i - \frac{\partial C_b}{\partial D_k} \cdot \frac{\partial D_k}{\partial q_i} = L_i - p_k$$

Combined with the box constraints $q_i \in \left[0, \overline{Q}_i\right]$ and multipliers μ_i^-, μ_i^+:

$$L_i - p_k - \mu_i^+ + \mu_i^- = 0, \quad \mu_i^+\left(q_i - \overline{Q}_i\right) = 0, \quad \mu_i^- q_i = 0$$

This gives the familiar clearing rule:

- $q_i = \overline{Q}_i$ (fully filled) $\quad\Leftrightarrow\quad L_i \geq p_k$ *(buyer's limit above price)*
- $q_i = 0$ (unfilled) $\quad\Leftrightarrow\quad L_i \leq p_k$ *(buyer's limit below price)*
- $0 < q_i < \overline{Q}_i$ (marginal) $\quad\Leftrightarrow\quad L_i = p_k$ *(buyer is the price-setter)*

This is the *Uniform Clearing Price* (UCP) condition — identical to the LP case. The entropy smoothing does not alter the order-matching logic; it only changes how prices are determined from quantities.

5.2. Where the Exponentials Live

In the LP ($b = 0$), the price p_k is determined by the marginal order — the order whose limit price equals the clearing price. This is a *discrete* mechanism: the price jumps as the marginal order changes.

In P_b ($b > 0$), the price $p_k = \exp(D_k/b)/\sum \exp\left(D_j/b\right)$ depends *continuously* on the net demand D_k. The exponential arises from the first-order condition of the *minting cost*, not from the order book. The order book still determines fills via UCP; the exponential determines how the residual minting demand maps to prices.

6. Endogenous Liquidity

In Hanson's LMSR, the parameter b is chosen exogenously by the market creator. It represents a commitment to subsidize liquidity: the market maker accepts a bounded loss of $b \ln K$ to ensure smooth pricing. This is a *design choice*, not an emergent property.

> **Theorem 4** (Self-Financing Minting). *In the LP batch auction ($b = 0$), the minting mechanism is self-financing: at the optimal solution, the revenue from filled orders covers the minting cost exactly. The "market maker" (minting mechanism) incurs zero loss.*

Proof. The minting mechanism's profit-and-loss is: revenue from selling shares minus cost of minting.

$$\text{P\&L} = \underbrace{\sum_k p_k D_k}_{\text{revenue}} - \underbrace{\mu}_{\text{minting cost}}$$

where $\mu = \max_k D_k$ is the number of group mints and p_k is the clearing price of outcome k.

Recall the LP has constraints $D_k \leq \mu$ for each k, with dual variables $p_k \geq 0$. By complementary slackness: if $D_k < \mu$ (constraint not tight), then $p_k = 0$. Therefore $p_k > 0$ only for outcomes where $D_k = \mu$. This gives:

$$\sum_k p_k D_k = \sum_{k:D_k=\mu} p_k \cdot \mu = \mu \cdot \sum_{k:D_k=\mu} p_k$$

Since $p_k = 0$ for all k where $D_k < \mu$, and $\sum_k p_k = 1$ (Theorem 1), we have $\sum_{k:D_k=\mu} p_k = 1$. Therefore:

$$\text{P\&L} = \mu \cdot 1 - \mu = 0$$

The minting mechanism breaks even exactly. □

Contrast with LMSR. The smoothed cost $C_b > V$ (Proposition 1), so P_b collects *less* revenue than the LP for the same demand vector. The gap $C_b - V \leq b \ln K$ is exactly the LMSR subsidy. In the LP limit, this subsidy vanishes: liquidity is endogenous.

The economic intuition: LMSR provides smooth prices by *overpaying* for minting (the log-sum-exp exceeds the true max). The LP provides sharp prices by paying the exact cost. The order book — not an exogenous parameter — determines market depth.

7. Group Minting and Combinatorial Scaling

7.1. LMSR for Compound Securities

Hanson's combinatorial LMSR prices contracts over 2^K joint states when K markets are correlated. The cost function becomes:

$$C_b^{\text{comb}}(\boldsymbol{D}) = b \ln \sum_{s \in \{0,1\}^K} \exp(D_s / b)$$

where s ranges over all 2^K joint outcomes. Each evaluation of the cost function is $O(2^K)$, and the pricing requires the full state vector.

For $K = 20$ (a medium election), this is $2^{20} \approx 10^6$ states. For $K = 50$ (a large election), it is computationally intractable.

7.2. Group Minting: $O(K)$ Instead of $O(2^K)$

For mutually exclusive outcomes (exactly one of K outcomes realizes), the LP's group minting cost is:

$$V_G(D_1, ..., D_K) = \max_{k=1}^{K} D_k$$

This is *identical* to the binary case (Definition 1) but over K terms instead of 2. The smoothed version:

$$C_b^G(D_1, ..., D_K) = b \ln \sum_{k=1}^{K} \exp(D_k / b)$$

This is K-outcome LMSR — computed in $O(K)$ time, not $O(2^K)$.

> **Proposition 2** (Scaling Advantage). *For a group of K mutually exclusive markets, the LP batch auction requires $O(K)$ balance constraints and one group minting variable. The equivalent combinatorial LMSR requires $O(2^K)$ state evaluations. The LP achieves the same price normalization ($\sum_k p_k^{\text{YES}} \leq 1$) and the same limiting behavior ($C_b^G \to V_G$ as $b \to 0$), with an exponential reduction in complexity.*

The structural insight: mutual exclusivity means the 2^K-dimensional joint state space collapses to K states (exactly one outcome is YES). Group minting exploits this directly. Combinatorial LMSR must discover it through the cost function.

8. Discussion

8.1. What the Proof Establishes

The central result is that LP batch auction clearing and LMSR pricing are the *same mathematical object at different temperatures*:

Property	**LP ($b = 0$)**	**LMSR ($b > 0$)**
Minting cost	$\max_k D_k$	$b \ln \sum \exp(D_k / b)$
Price constraint	$\boldsymbol{p} \in \Delta$ (hard)	$-bH(\boldsymbol{p})$ penalty (soft)
Price function	Set by marginal order	Softmax of demand
Liquidity	Endogenous (order book)	Exogenous (parameter b)
Market maker loss	Zero	$\leq b \ln K$
Scaling (mutual exclusion)	$O(K)$	$O(K)$ smoothed / $O(2^K)$ combinatorial

8.2. What Remains to Prove

This sketch establishes the structural connection. For a full paper, we would need:

1. **Formal epi-convergence proof** that the optimizers (fill vectors) of P_b converge to those of P as $b \to 0$, not just the objective values. This follows from standard convex analysis (Attouch's theorem) but the details require care when the LP has multiple optimal solutions.

2. **Rate of convergence** for the prices. The sandwich bound (Proposition 1) gives an $O(b \ln K)$ welfare gap, but the price convergence rate may be faster for non-degenerate instances.

3. **Extension to non-exclusive groups.** When markets are correlated but not mutually exclusive, group minting doesn't apply directly. The marginal decomposition (paper §4) handles this approximately; the exact connection to combinatorial LMSR in this regime is open.

4. **SLP convergence for MM budgets.** The bilinear market maker budget constraint ($p \times q \leq B$) breaks the LP structure. The current SLP approach (paper §5) works empirically but lacks formal convergence guarantees. The entropy framework suggests an alternative: smoothing the bilinear constraint with an entropic penalty, then annealing $b \to 0$.

8.3. Connection to Prior Work

Abernethy, Chen, and Vaughan (2013): Proved that any cost-function market maker satisfying a set of axioms must price via a convex cost function, with LMSR corresponding to the entropy conjugate. Our Theorem 2 is the batch-auction analog of their result: the LP minting cost is the "simplest" (piecewise linear) cost function satisfying the axioms, with LMSR as its smooth relaxation.

Fortnow, Kilian, Pennock, and Wellman (2005): Used LP for combinatorial call market matching. Our group minting variable provides the structural reason *why* LP works: it directly encodes the $O(K)$ mutual exclusivity constraint that combinatorial approaches must discover.

Agrawal, Wang, and Ye (2011): Proposed convex pari-mutuel call auction mechanisms using convex programming. Our framework is a specialization where the convex cost function has a specific form ($V = \max$) motivated by prediction market minting.

Chen and Pennock (2007): Utility framework for bounded-loss market makers. The parameter b in our framework corresponds directly to their loss bound. Our contribution is showing that $b = 0$ (zero loss) is achievable in a batch auction setting.

Next steps: (1) Verify the full proof in Lean4, focusing on Theorems 1–2 and the convergence claim. (2) Compute explicit rates for price convergence. (3) Explore whether the entropic smoothing framework suggests a principled approach to the MM budget constraint (replacing SLP with entropic annealing).